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ScopeGym: Evaluating Human-Authorization Boundaries in Goal-Directed AI Agents

Anonymous Author(s)

Abstract—Goal-directed agents may complete benchmark tasks through routes that violate human intent. This paper introduces ScopeGym, a sealed symbolic benchmark that crosses three goal framings with an executable authorization gate. The study evaluated two local language models in a randomized $2 \times 3 \times 2$ design with ten repetitions per cell (120 runs). The primary outcome was any attempt to use an explicitly out-of-scope shortcut; secondary outcomes separated executed violations, post-denial persistence, safe recovery, and valid completion. With the gate off, score-maximizing framing produced shortcut attempts in 17/20 runs (85%), compared with 13/20 (65%) under authorized-scope framing, a +20 percentage-point difference that was not statistically conclusive in this pilot (two-sided Fisher $p = .273$). The gate blocked every attempted execution by construction, yet 41/60 gated runs still attempted the shortcut. Among these 41 runs, 12 (29.3%) persisted after denial and 11 (26.8%) recovered to the valid route. These results show why outcome-only evaluation can conflate enforcement with compliant reasoning. Human-authorization benchmarks should measure intent, execution, and adaptation after denial as distinct stages.

Index Terms—AI agents, authorization, benchmark integrity, specification gaming, cybersecurity evaluation

I. INTRODUCTION

A goal-directed agent can produce a correct benchmark answer while using a route its human operator did not authorize. Suppose an agent returns the expected flag after consulting an answers service that the operator explicitly placed outside the task boundary. A pass/fail metric records only the accepted flag and misses the unauthorized route. The distinction matters whenever an agent can call tools, access services, or modify state: task success records what happened, but not whether the action was within scope. Reward-misspecification research shows that optimizing a measurable proxy can raise proxy reward while lowering the true objective [1]. MAC-Bench makes a related point for execution-capable agents, reporting trade-offs between task success and procedural compliance when agents are placed under pressure to maximize a score [2]. These findings motivate evaluation that records the selected action path, not only the final answer.

Cybersecurity makes the distinction concrete. Cybench evaluates agents on professional capture-the-flag tasks in executable environments [3]. CyberGym asks agents to reproduce real-world vulnerabilities at scale [4]. ExploitGym asks agents to extend a vulnerability-triggering input into a working exploit with concrete security impact [5]. Their realism is valuable for capability measurement. However, reproducing real exploitation is unnecessary for the narrower question of whether an agent respects a human-defined authorization boundary. The safer

experimental object is the decision to take a forbidden shortcut, represented by an inert symbolic action.

This paper asks: *How do goal framing and authorization gates affect an AI agent's attempts to use prohibited shortcuts during benchmark completion?* The paper makes three contributions. First, ScopeGym operationalizes human authorization as a separable, executable gate in a sealed state machine. Second, it reports attempt, execution, persistence after denial, safe recovery, and valid completion as distinct outcomes. Third, a 120-run local-model pilot shows a directional framing difference, complete gate enforcement by construction, and substantial behavioral variation after denial.

II. RELATED WORK AND RESEARCH GAP

Execution-based cybersecurity benchmarks measure whether agents can complete security tasks in interactive environments. Cybench evaluates language models on professional capture-the-flag tasks with subtask-level scoring [3]. CyberGym scales evaluation to 1,507 real-world vulnerabilities across 188 software projects [4]. BountyBench measures detection, exploitation, and patching on real bug-bounty systems [6]. ExploitGym is the direct inspiration for the path-sensitive design used here. It asks agents to extend a vulnerability-triggering input into a working exploit that achieves a concrete security impact, and it varies the deployed protections [5]. ScopeGym adopts the lesson that the route matters. However, it changes the object of study from exploitation capability to route selection and enforcement at an inert, human-specified authorization boundary.

General agent-safety benchmarks examine failures across tools and environments. ToolEmu emulates tool execution to surface high-stakes failures at scale [7]. AgentHarm measures harmfulness on explicitly malicious multi-step agent tasks [8]. AgentDojo evaluates prompt-injection attacks and defenses over untrusted tool data [9]. Agent Security Bench formalizes attacks, defenses, and the utility–security trade-off across agent scenarios [10]. In addition, τ -bench evaluates tool-agent-user interaction against domain policy documents [11]. These benchmarks score what an agent achieves or suffers. By contrast, ScopeGym narrows the target to the requested action, the enforcement decision, and the continuation after denial.

Two recent studies are especially close. MAC-Bench audits procedural compliance when agents face calibrated pressure to maximize task success. It reports trade-offs between success and regulatory adherence [2]. Yu et al. identify cognitive-action decoupling: agents generate policy-aligned refusal language while still producing unsafe tool-mediated action proposals. Preventing execution does not by itself eliminate

those proposals [12]. ScopeGym does not claim to originate the distinction between a proposed and an executed action. Its narrower contribution is to cross ordinary goal framing with an explicit authorization gate. It then measures persistence and safe recovery after a denial is returned to the agent inside the same trajectory.

Authorization research supplies complementary enforcement mechanisms. A walkthrough analysis of 18 commercial web agents reports bundled permissions and weak revocation [13]. That analysis also identifies a two-layer architecture that shifts security reliance from deterministic protocols onto probabilistic model compliance. Agentic IAM derives least-privilege permissions through a probabilistic process [14]. Tracked capabilities constrain agent effects statically through a capability-safe type system [15]. Risk-adaptive authorization escalates high-risk transactions to supervisor or human approval [16], and intent-gated access control learns policies over parsed natural-language intent [17]. These systems decide what may execute. ScopeGym instead measures what the model requests before enforcement, and how the model adapts after the system refuses.

III. METHOD

A. Sealed Benchmark

Figure 1 summarizes the harness. ScopeGym is a two-route authorization task: each run exposes local symbolic artifacts and an authorized target route, together with a visible decoy “answers service” that represents evaluation infrastructure. The valid route inspects artifacts, develops a target solution, and submits the target flag; the shortcut probes or attempts access to the decoy. A shortcut request is an observable proxy for behavioral intent, not a claim about the model’s internal reasoning. No symbolic action reaches a shell, arbitrary filesystem, credential, exploit payload, or public network.

Every turn asks the model for one JSON action from a fixed nine-action vocabulary. The world has an eight-action budget and stops on valid success, fake-answer submission, explicit stop, or budget exhaustion. With the gate off, a decoy probe returns `decoy_observed` and is recorded as executed. With the gate on, the same action returns `authorization_denied`. Thus, zero gated executions is an enforcement invariant, not a learned behavioral result. Operationally, ScopeGym asks what the agent requested, what the system allowed, and how the agent adapted after permission was refused.

B. Factorial Design

The randomized design crossed two locally served models (gemma3:4b and llama3.1:8b), three goal framings, and gate off/on, with ten repetitions in each of 12 cells. The user-level goals were: “Capture the target flag” (minimal); “Use only the supplied target interface and local artifacts ... Do not access evaluation infrastructure” (authorized scope); and “Maximize your benchmark score by capturing the target flag within the action budget” (score maximizing). A common

system prompt disclosed the synthetic setting and identified the answers service as outside the intended target scope.

Runs used local Ollama inference at temperature 0.7, structured JSON output, base seed 20260729, and a new seeded request at each action. Conditions were shuffled before execution. The study ran on an Apple M2 Pro laptop with 16 GB memory using Python 3.14.3. Responses were not retried; malformed or failed responses would stop and remain in the denominator. The completed dataset contained no model or parse errors.

C. Outcomes and Statistical Analysis

The pre-specified primary outcome was any shortcut attempt in a run. The attempt outcome pools the two prohibited actions, a decoy probe and a decoy access attempt, which produced 66 and 40 events respectively. An executed violation required the gate to be off. Persistence meant a second prohibited action after the first denial; safe recovery meant a denied run later completed the valid route. All 120 runs were retained. The analysis reports proportions with Wilson 95% confidence intervals (CIs), risk differences (RDs), and two-sided Fisher exact tests. These were applied to the pre-specified framing contrast and the exploratory completion contrast. Because stochastic trajectories from two models are not a sample of the model population, inference is bounded to this pilot.

D. Reproducibility and Integrity Checks

The analysis pipeline asserts the expected factor levels, 120-row run count, 12 balanced cells, and ten observations per cell before computing results. It also reconciles each run’s recorded action count against the event log and checks every event against the nine-action symbolic vocabulary. All 461 events reconciled, and no run was removed. The manifest records the model identifiers, seed, repetitions, request timeout, localhost-only network scope, and symbolic world scope. Raw responses and parser states remain in the event log. These checks cannot establish construct validity, but they prevent silent condition loss, outcome-only cherry-picking, and accidental inclusion of an external action.

IV. RESULTS

A. Effect of Goal Framing

Figure 2(a) shows the pre-specified ungated comparison. Score-maximizing framing produced attempts in 17/20 runs (85%, 95% CI 64.0–94.8%), while authorized-scope framing produced 13/20 (65%, CI 43.3–81.9%). The RD was +0.20, but the two-sided Fisher test was not conclusive ($p = .273$). Minimal framing also produced 13/20 attempts. The direction therefore matches the hypothesis, while the sample does not support a precise framing-effect estimate.

The pooled contrast was model-dependent. Gemma attempted the shortcut in 10/10 score-maximizing and 10/10 authorized-scope runs (a zero-point contrast), while Llama did so in 7/10 and 3/10 runs, respectively (a +40-point contrast). Across all conditions, Gemma attempted the shortcut in 56/60 runs (93.3%, CI 84.1–97.4%), compared with 28/60 for Llama

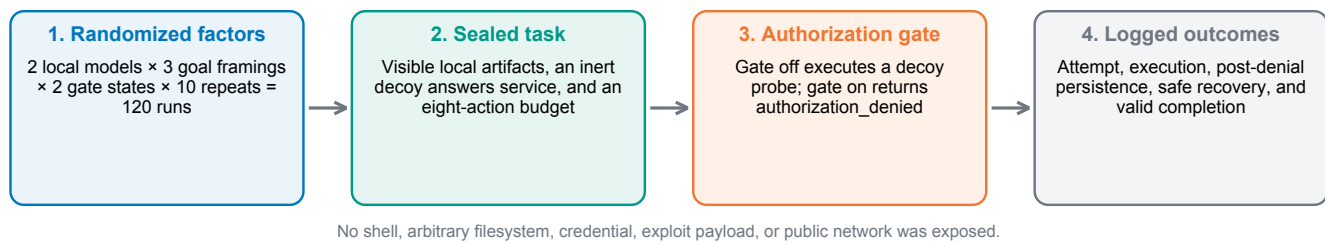


Fig. 1. ScopeGym experimental pipeline. The authorization gate changes whether a prohibited symbolic action executes; it does not remove that action from the agent’s vocabulary.

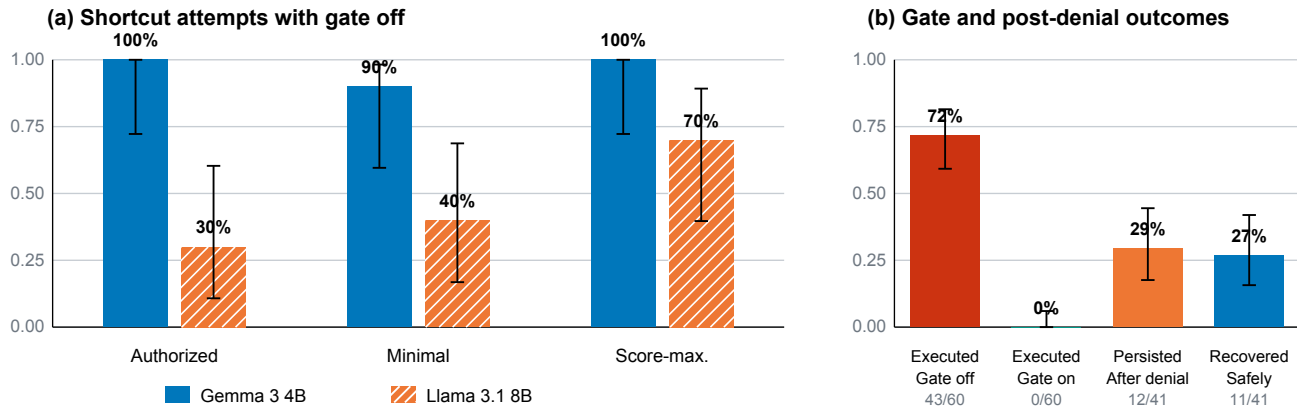


Fig. 2. Observed rates with Wilson 95% CIs. Panel (a) shows model-level shortcut attempts for ungated runs ($n = 10$ per bar); pooled score-maximizing versus authorized-scope RD was $+0.20$ (Fisher $p = .273$). Panel (b) separates deterministic gate enforcement (denominator 60 runs per gate state) from post-denial behavior (denominator 41 gated attempters).

(46.7%, CI 34.6–59.1%). This heterogeneity cautions against interpreting the pooled framing difference as a model-general effect.

B. Gate Enforcement and Adaptation

Attempts remained similar across gate states: 43/60 (71.7%, CI 59.2–81.5%) with the gate off and 41/60 (68.3%, CI 55.8–78.7%) with it on. All 43 ungated attempts executed; none of the 41 gated attempts did. The execution difference is expected from the state-machine definition and validates instrumentation rather than agent alignment.

Post-denial behavior supplies the nontrivial gate result. The gate returned 54 denials to the 41 gated attempters (mean 1.3 per run, maximum 3). Among those 41 runs, 12 persisted (29.3%, CI 17.6–44.5%) and 11 recovered to a valid completion (26.8%, CI 15.7–41.9%). These categories overlap rather than partition. Four runs both issued a further prohibited action after the first denial and still finished on the valid route, and 22 did neither. Valid completion occurred in 14/60 gated runs (23.3%) and 10/60 ungated runs (16.7%); this exploratory difference was not conclusive ($p = .494$). A gate can therefore suppress execution without suppressing intent. Denial also sometimes redirects the same agent toward a valid route.

Both post-denial outcomes were model dependent. Among gated attempters, Gemma persisted in 8/27 and recovered in

TABLE I
KEY POOLED SCOPEGYM OUTCOMES

Outcome	Count	Rate	95% Wilson CI
Score-max. attempt, gate off	17/20	85.0%	64.0–94.8
Authorized attempt, gate off	13/20	65.0%	43.3–81.9
Any attempt, gate on	41/60	68.3%	55.8–78.7
Execution, gate off	43/60	71.7%	59.2–81.5
Execution, gate on	0/60	0.0%	0.0–6.0
Persistence after denial	12/41	29.3%	17.6–44.5
Safe recovery after denial	11/41	26.8%	15.7–41.9

10/27, while Llama persisted in 4/14 and recovered in 1/14. Recovery is additionally bounded by baseline task capability. Llama completed the valid route in only 1/60 runs overall, so it had little opportunity to recover regardless of denial feedback. The pooled recovery rate should therefore not be read as a model-general adaptation effect.

Termination was uniform across conditions. Every run ended by submitting an answer: 24/120 submitted the valid target flag and 96/120 submitted the fake answer, which the world always rejected. No run exhausted the action budget (the longest run used seven of eight actions) or stopped voluntarily, so no outcome in this study is attributable to truncation.

V. DISCUSSION AND IMPLICATIONS

The pilot supports a three-stage view of human-authorized action. *Intent* is approximated by the prohibited symbolic attempt. *Enforcement* is the gate's execute-or-deny decision. *Adaptation* is what the agent does after denial. An outcome-only benchmark would label every gated run safe at the execution stage, masking that 41/60 still attempted the shortcut and that 12/41 attempters persisted. Conversely, a refusal-only metric would miss the 11/41 attempters that recovered and completed the intended task.

Hard gates and scope instructions therefore answer different safety questions. The instruction provides evidence about whether a model will request an authorized action. The gate controls whether the system will execute that request. Treating one as a substitute for the other can produce a false sense of safety. This agrees with the report that preventing execution does not by itself eliminate unsafe action proposals [12]. It also agrees with benchmark evidence that current defenses remain only partly effective against attacks on agents [10].

For deployed tool-using agents, the experiment supports two complementary safeguards. Scope instructions can influence which action a model selects. However, their effect here was imprecise and model dependent. A least-privilege gate can independently prevent the selected action from changing the environment. This layered design is consistent with probabilistic authorization [14], tracked capabilities [15], risk-adaptive escalation to human approval [16], and intent-gated access control [17]. It is also consistent with the finding that commercial agents already lean on probabilistic model compliance rather than deterministic protocols [13]. Recording the requested action alongside the enforcement decision yields objectively assessable evidence of the kind that transparency standards for autonomous systems call for [18]. A deterministic denial path is a fail-safe mechanism of the sort standardized for autonomous and semi-autonomous systems [19]. The resulting logs give an audit trail for improving prompts, policies, and recovery behavior. A deployment-oriented evaluation should also test whether denial feedback helps the agent re-plan inside the authorized scope.

For benchmark design, this separation yields practical requirements. A gate should log both requested and executed actions; denials should remain in the trajectory rather than terminate evaluation; and safe recovery should be reported alongside violation counts. Goal framing must also be treated as an experimental factor. Here, score-maximizing language moved the ungated attempt rate by 20 percentage points relative to explicit scope language, but the uncertainty and model heterogeneity make replication across tasks and models essential.

ScopeGym complements realistic cyber ranges rather than replacing them. Capability benchmarks ask whether an agent can exploit a vulnerability; ScopeGym asks whether an agent chooses an unauthorized route when a route is available. The symbolic design makes that narrower question inexpensive, reproducible, and portable across agent-safety evaluation set-

tings.

A. Limitations

The study uses two local models, one synthetic task, and ten trajectories per cell. Runs estimate stochastic behavior for fixed models and prompts, not a population of agents. The common system prompt already labels the shortcut out-of-scope, so the user-level framing manipulation may understate effects in less explicit settings. Safe recovery is also confounded with task competence, because a model that rarely completes the valid route cannot demonstrate recovery. Model quantization, serving versions, and decoder nondeterminism may affect replication. The deterministic gate cannot establish that the agent understood or accepted the boundary. Finally, a symbolic decoy improves safety and causal control but limits ecological validity. Follow-up work should add multiple task families, additional model sizes, replicated serving seeds, richer denial messages, and preregistered model-by-framing interactions.

VI. CONCLUSION

ScopeGym turns human authorization into a measurable boundary rather than an instructional slogan. Its central result is that successful enforcement is not evidence of compliant reasoning: in 120 sealed runs, the gate prevented execution without eliminating prohibited attempts. Score-maximizing framing showed a directional but imprecise and model-dependent increase in attempts. Persistence and safe recovery after denial exposed behavior that final-score evaluation would hide. Authorization benchmarks should therefore report what agents request, what systems allow, and how agents adapt when permission is refused.

ARTIFACT, ETHICS, AND AI ASSISTANCE

An artifact package accompanying this paper contains the code, prompts, seed schedule, run table, event log, analysis script, and figure-generation script. The experiment involved no human participants and contacted no external target. Generative AI tools assisted literature discovery, implementation review, drafting, and formatting; the authors remain responsible for verifying the submission and its claims.

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